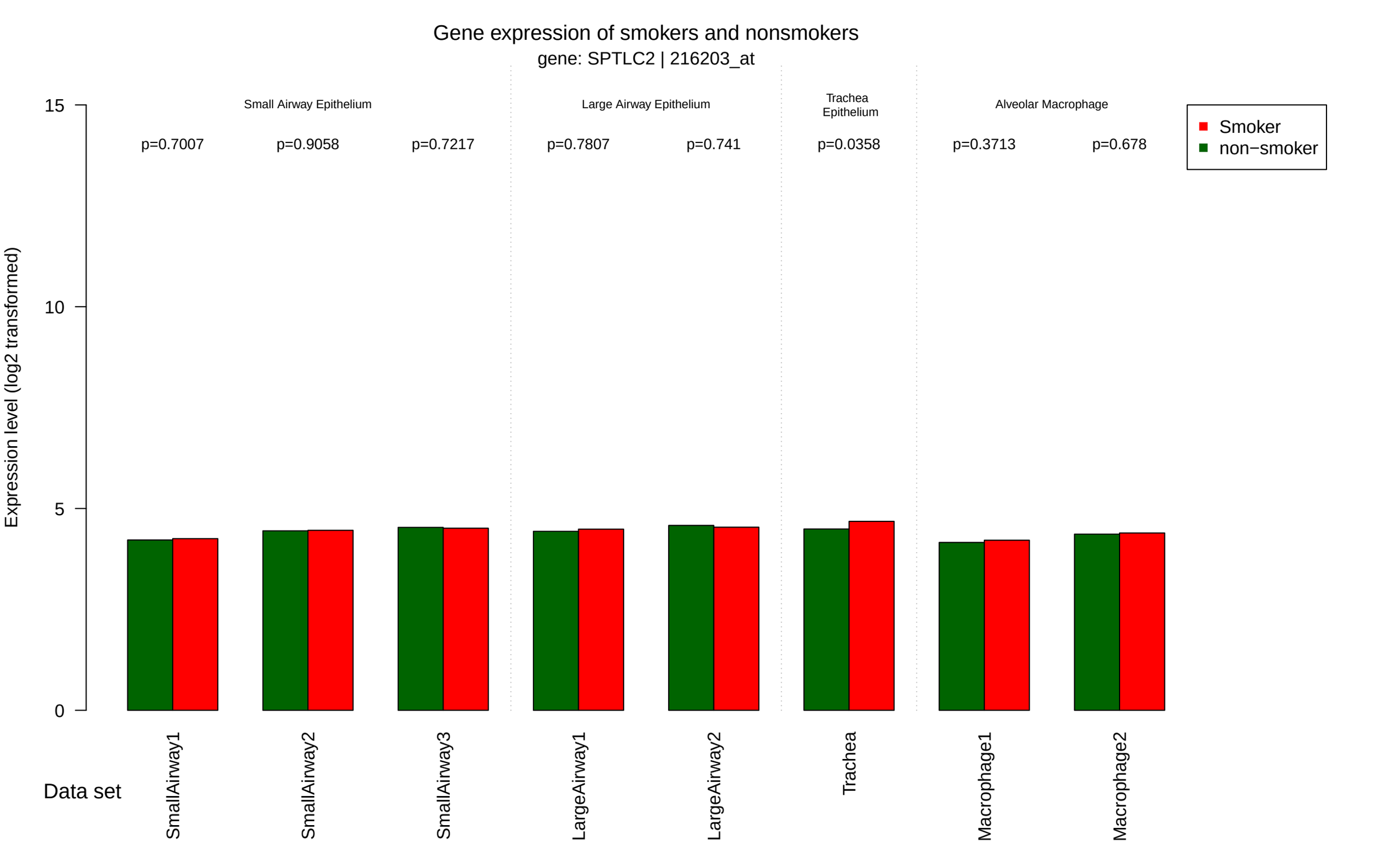
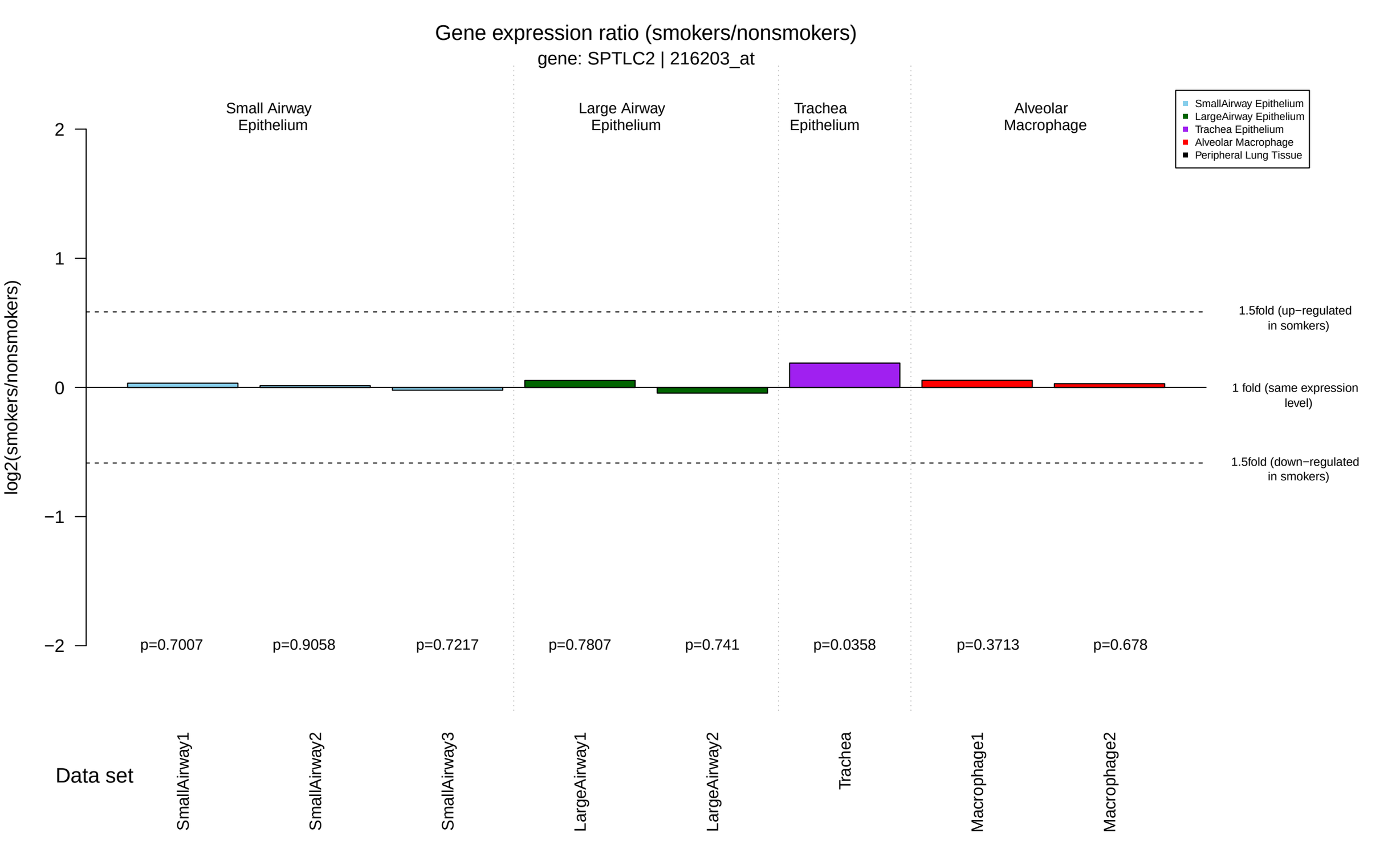
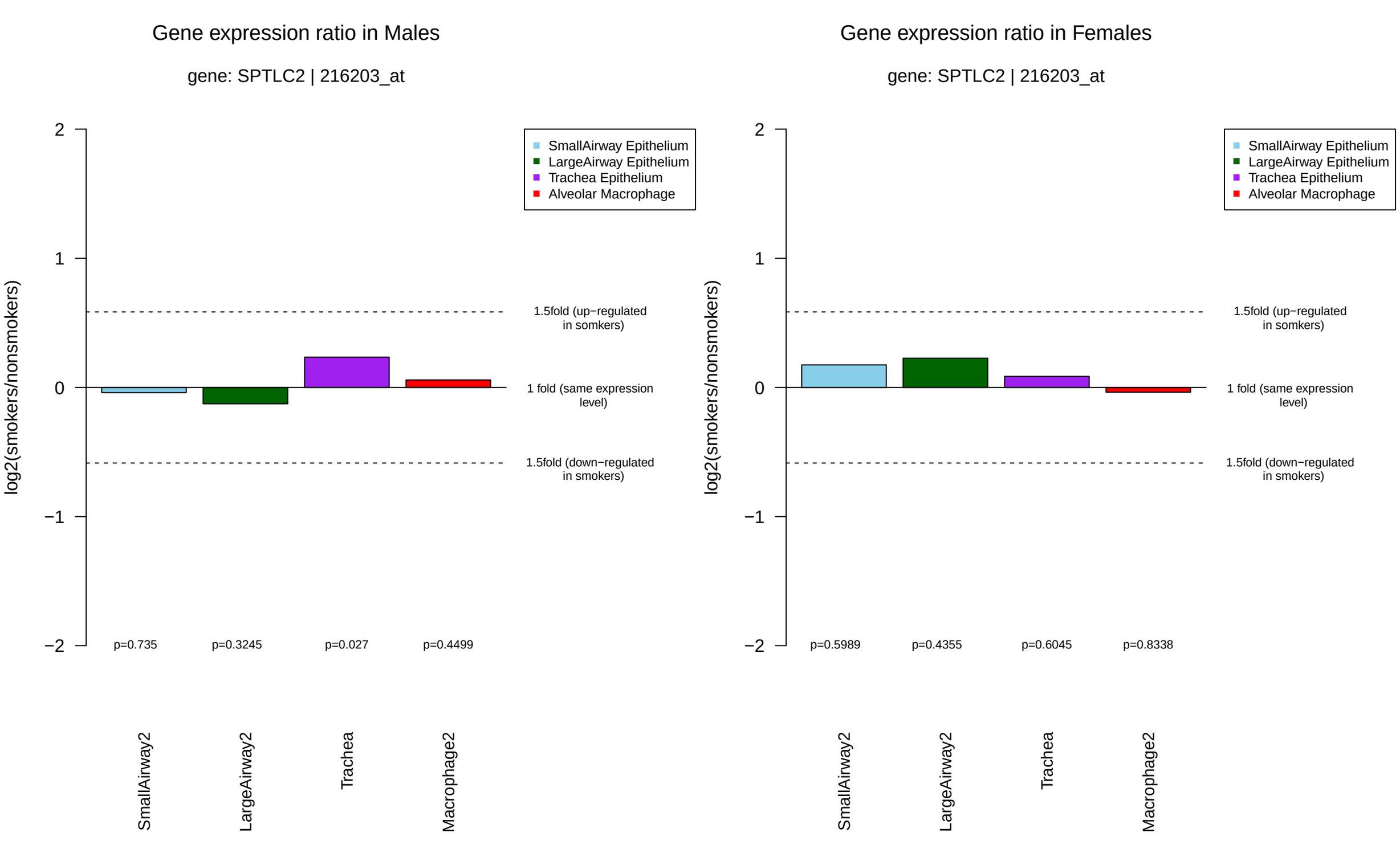
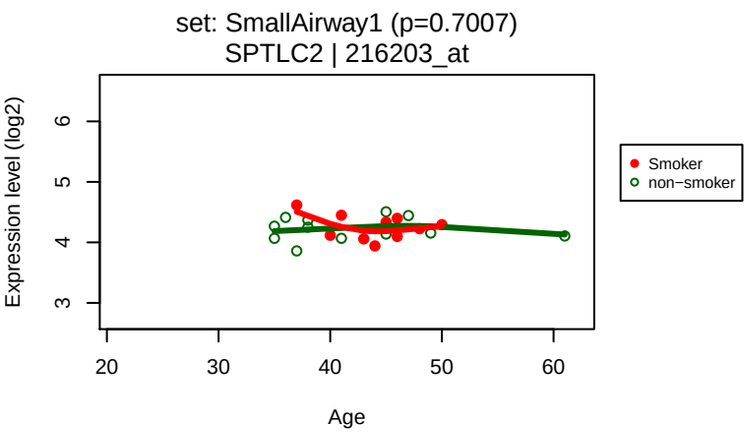
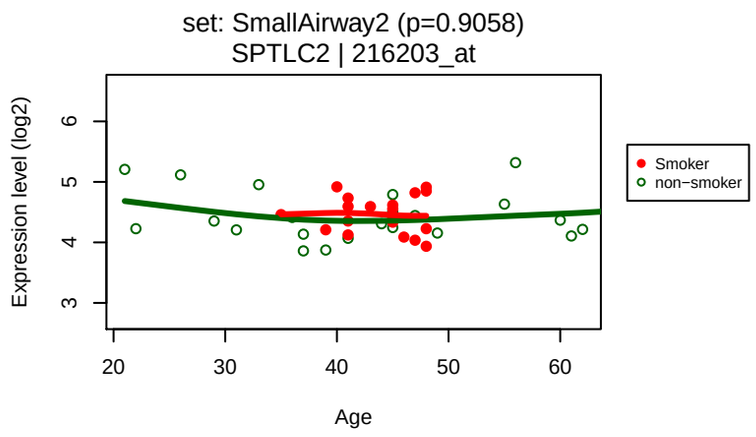
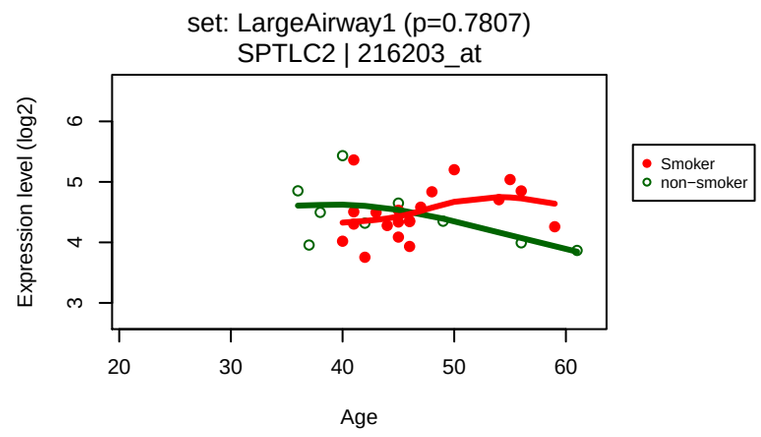

Gene expression of smokers and nonsmokers
gene: SPTLC2 | 216203_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7007
p=0.9058
p=0.7217
p=0.7807
p=0.741
p=0.0358
p=0.3713
p=0.678
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

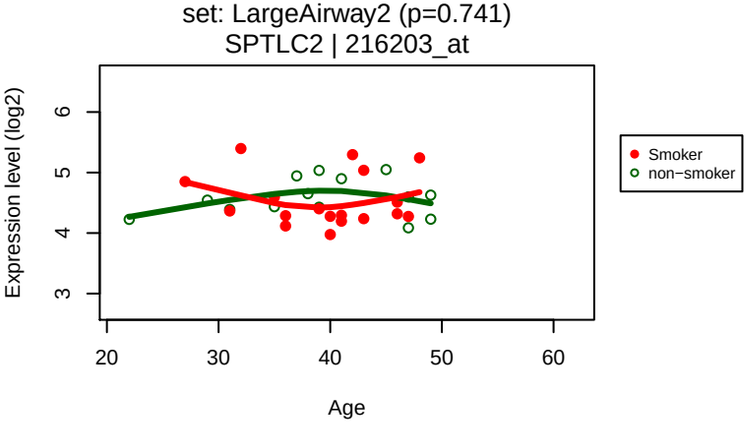

Gene expression ratio (smokers/nonsmokers)
gene: SPTLC2 | 216203_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7007
p=0.9058
p=0.7217
p=0.7807
p=0.741
p=0.0358
p=0.3713
p=0.678
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

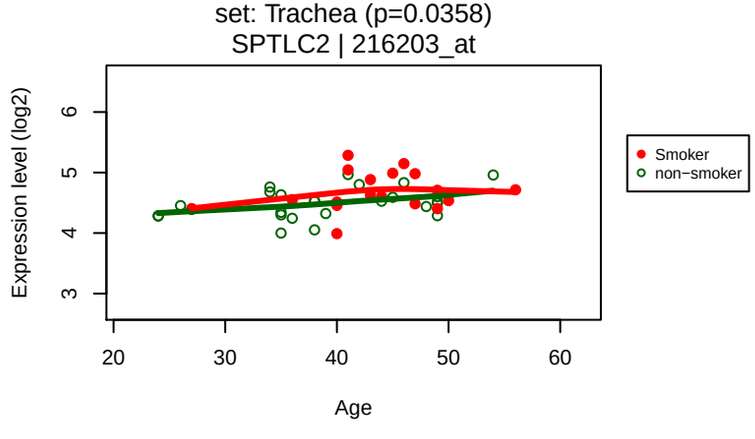

Gene expression ratio in Males
gene: SPTLC2 | 216203_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.735
p=0.3245
p=0.027
p=0.4499
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPTLC2 | 216203_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5989
p=0.4355
p=0.6045
p=0.8338
SmallAirway2
LargeAirway2
Trachea
Macrophage2

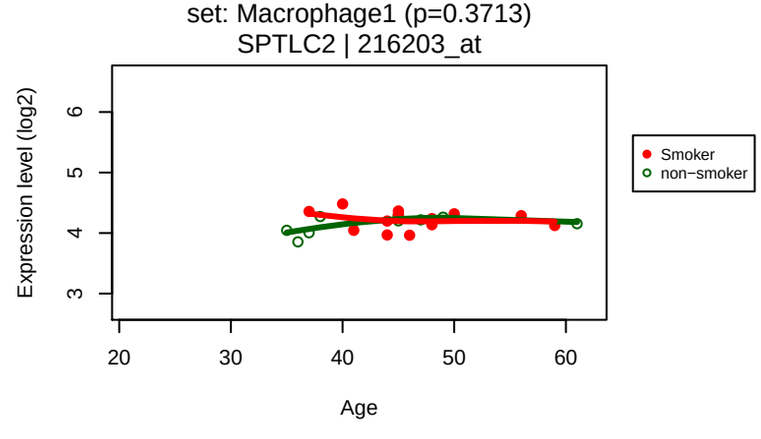

set: SmallAirway1 (p=0.7007)
SPTLC2 | 216203_at
Expression level (log2)
Age
Smoker
non−smoker

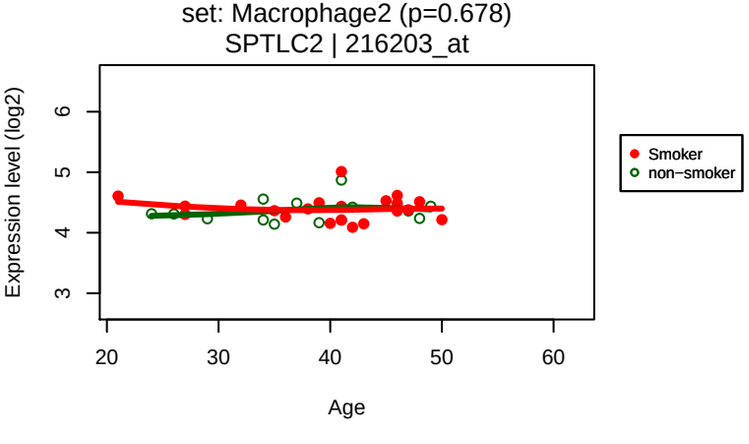

set: SmallAirway2 (p=0.9058)
SPTLC2 | 216203_at
Expression level (log2)
Age
Smoker
non−smoker

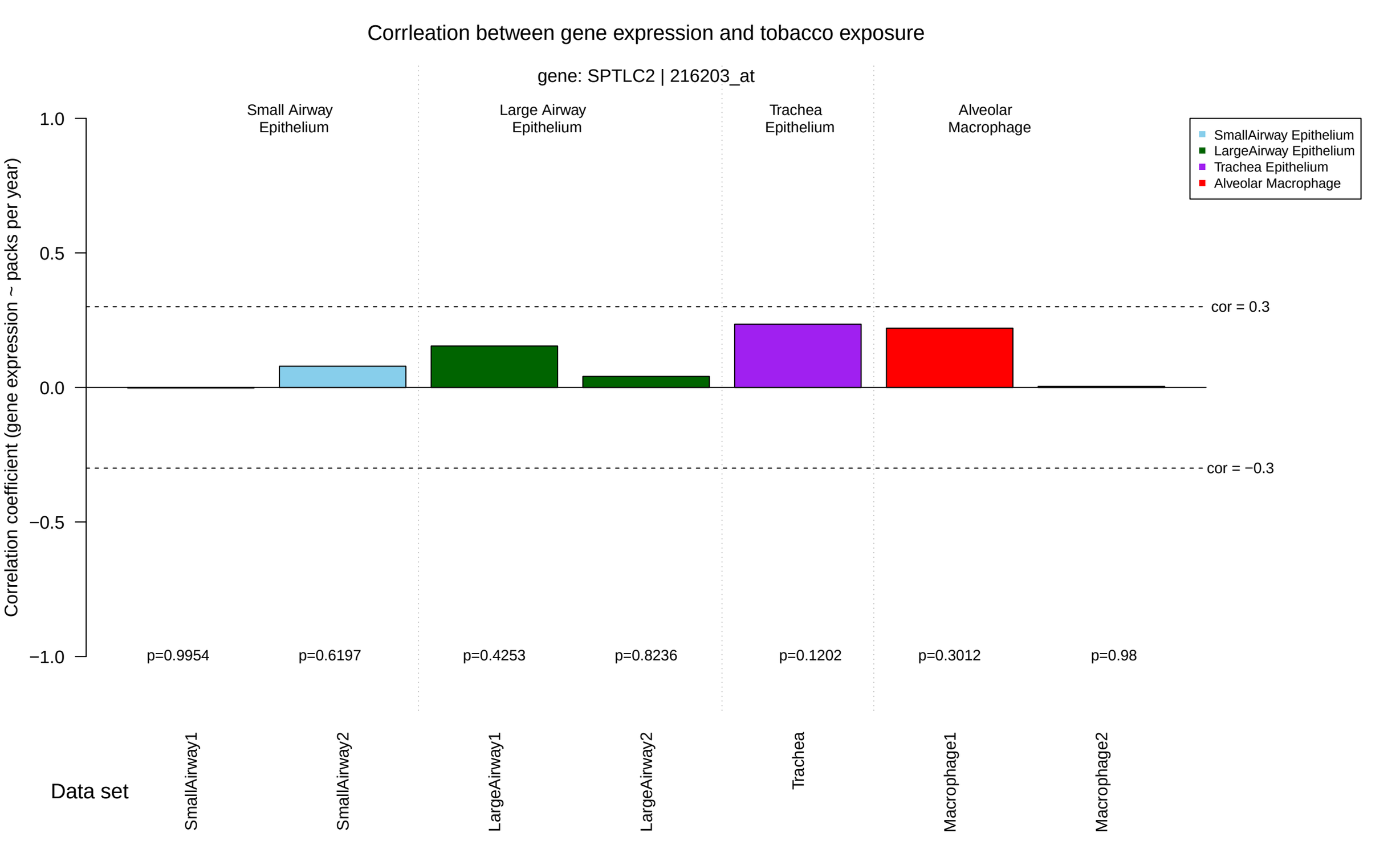

set: LargeAirway1 (p=0.7807)
SPTLC2 | 216203_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.741)
SPTLC2 | 216203_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0358)
SPTLC2 | 216203_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3713)
SPTLC2 | 216203_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.678)
SPTLC2 | 216203_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPTLC2 | 216203_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9954
p=0.6197
p=0.4253
p=0.8236
p=0.1202
p=0.3012
p=0.98
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2